7.3. ESSENTIAL PARTS OF AN OSCILLATOR

Figure 7.3 shows the block diagram of an oscillator. Its essential components are:

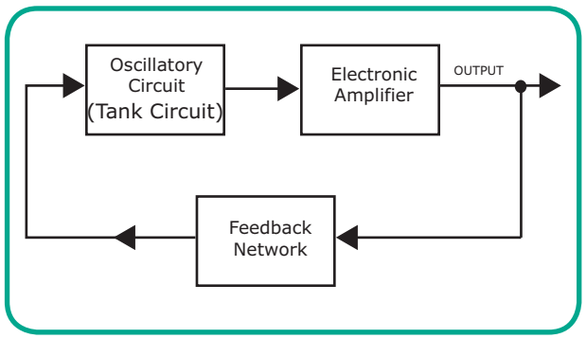


Figure 7.3 **Block Diagram of an Oscillator**

Tank Circuit: It consists of inductor or coil (L) and capacitor (C). The frequency of oscillation depends upon the values of inductance of the coil and capacitance of the capacitor.

Transistor Amplifier: The transistor amplifier receives DC power from the battery and changes it into AC power for supplying to the tank circuit. The main function of the amplifier is to amplify the generated oscillation from the tank circuit.

Feedback Circuit: This circuit provides positive feedback to the oscillator. It gives a part of amplifier output to the tank circuit in correct phase to make oscillation as undamped (constant amplitude).

7.4. FEEDBACK IN OSCILLATOR

In feedback, a part of the output signal is fedback to the amplifier input in such a way that the feedback signal re-generates, re-amplifies and sustain the feedback to maintain a constant output Signal.

Commonly an oscillator is constructed from an amplifier that has part of its output signal feedback to the input. This is done in such a way to keep the amplifier producing signal without the need for any external signal input as shown in the Figure 7.4. Here, the DC supply is converted into AC signal.

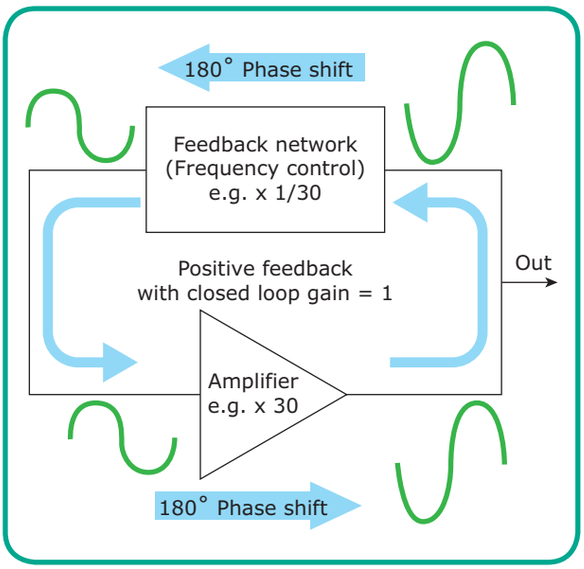


Figure 7.4 **Positive Feedback**

7.4.1. Positive Feedback

The feedback in the amplifier section of an oscillator must be positive feedback. Here a fraction of the amplifier output signal is feedback as input. Note that the feedback signal is in phase with the input signal. As a result, the amplitude of the signal is increased.

For example, common emitter amplifier creates a phase shift of 180° between its input and output. Similarly, the positive feedback loop also produce a 180° phase shift in the signal feedback from output to input in order to provide positive feedback.

The result of small amount of positive feedback in amplifiers results in

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